

these “integrated cells” within a new paradigm of “the cell alignment.” Similarly, the world’s economic system includes networks of trade, financial exchanges, and investments, which thus far have been studied individually (3, 5). But a complete understanding of the origins of wealth, crises, and economic recoveries can only come from aligning and collectively analyzing all of these layers of networked economic and geopolitical data. Likewise, climatic measurements are captured by various

## “Holistic analyses of our interconnected world call for conceptual and methodological paradigm shifts.”

network types encoding the relationships between climatic elements across geographical regions (e.g., wind speed, atmospheric pressure, and temperature) (4), and holistic, data-aligned analyses may help to explain this complex, dynamic system and better predict the effects of human-caused alterations. Mathematical formalisms capable of capturing the intricacies of higher-order organization of the data, along with the algorithms to compute and extract information from those formalisms, should be developed and applied (15). Extending the framework of Benson *et al.* to finding higher-order structures within these integrated and aligned data systems may be a way forward. Computational issues remain to be addressed, arising from large sizes, complexity, heterogeneity, noisiness, and different time and space scales of the data. ■

### REFERENCES AND NOTES

1. M. R. Garey, D. S. Johnson, *Computers and Intractability: A Guide to the Theory of NP-Completeness* (Freeman, New York, 1979).
2. A. R. Benson *et al.*, *Science* **353**, 163 (2016).
3. O. N. Yaveroğlu *et al.*, *Sci. Rep.* **4**, 4547 (2014).
4. K. Steinhäuser, A. A. Tsonis, *Clim. Dyn.* **42**, 1665 (2014).
5. P. Glasserman, H. P. Young, *J. Bank. Financ.* **50**, 383 (2015).
6. R. Sharan *et al.*, *Mol. Syst. Biol.* **3**, 1 (2007).
7. K. Mitra *et al.*, *Nat. Rev. Genet.* **14**, 719 (2013).
8. A. L. Barabási *et al.*, *Nat. Rev. Genet.* **12**, 1 (2011).
9. J. Menche *et al.*, *Science* **347**, 6224 (2015).
10. M. Žitnik *et al.*, *Sci. Rep.* **3**, 3202 (2013).
11. V. Gligorijević *et al.*, *Proteomics* **16**, 741 (2016).
12. S. M. Strittmatter, *Nat. Med.* **20**, 590 (2014).
13. D. Davis *et al.*, *Bioinformatics* **31**, 1632 (2015).
14. J. Alfoldi, K. Lindblad-Toh, *Genome Res.* **23**, 1063 (2013).
15. S. Boccaletti *et al.*, *Phys. Rep.* **544**, 1 (2014).

### ACKNOWLEDGMENTS

This work was supported by the European Research Council Starting Independent Researcher Grant (278212), the National Science Foundation Cyber-Enabled Discovery and Innovation Program (OIA-1028394), the Slovenian Research Agency (ARRS) Project (J1-5454), and the Serbian Ministry of Education and Science Project (III44006).

10.1126/science.aah3449

### ENGINEERING

# Solar-powering the Internet of Things

Photovoltaics can help to power remote sensors and controllers at the edge of the Internet

By Richard Haight, Wilfried Haensch, Daniel Friedman

The Internet connects billions of computational platforms of various sizes, from supercomputers to smart phones. However, the same types of data transmission can connect computational resources to much simpler sensors “at the edge of the net” that collect, analyze, and transmit data, as well as controllers that receive instructions. Devices deployed in the environment, homes and offices, and even our bodies would expand the number of connected devices to the trillions. This “Internet of Things” (IoT) underlies the vision of smart homes and buildings that could sense and transmit their status and respond appropriately (1), or track and report on the state of objects (vehicles, goods, or even animals) in the environment. However, the practical implementation of the IoT has been relatively slow, in part because all of these edge devices must draw electrical power from their local environment. We analyze the use of photovoltaics (PV) to power devices and help bring the IoT to fruition.

Wide-scale deployment of devices to remote or inaccessible areas while providing operational power in the absence of wires would require harvesting of available energy to ensure long-term operation. Indeed, one barrier to making an existing building “smart” is the cost of installing wired devices, so devices simply mounted to walls and powered by ambient indoor lighting are attractive for lowering installation costs. Whatever the specific form of energy used (alternatives to PV include kinetic, thermal, or wind energy), it must have sufficient energy density and reliability. The power requirements are dictated by the three components that process, sense, and communicate data. An excellent example of a system at the far edge of the net is the ultralow-power chip developed at the University of Michigan that incorporates much of the required functionality and energy-harvesting infrastructure (2).

An ultralow-power processor that might, for example, be used in a smart phone (3) would consume ~85  $\mu$ W of power when operating at a clock frequency of 1 MHz. Active power consumption in a processor is given by  $CV^2F$ , where  $C$  is the capacitance,  $V$  is the operating voltage of the processor, and  $F$  is the clock frequency. Reducing the operating voltage has the biggest benefit, and energy-efficient processors have been demonstrated to run with a supply voltage of <400 mV at 0.83 MHz (4). The time constant of the systems monitored is often long (milliseconds to minutes), so the required clock frequency can be well below 1 MHz. Intermittent operation—where the processor can be woken to take a measurement, store data, and later transmit it to a receiving station—would save appreciable power. Hence, ultralow-power operation of an edge device is both required and achievable.

Sensors are another area where low power is critical. As an example, consider an electrochemical sensor (5) used to monitor concentrations of toxic gases or other air-dispersed pollutants. A typical sensor might consist of a miniature electrochemical cell that draws 100 to 500  $\mu$ A of current at 1 to 2 V. The total power consumed depends on whether it operates continuously (as for a carbon monoxide monitor) or periodically (an outdoor ozone monitor for air quality). Typical power requirements for most sensors can range from 100  $\mu$ W to >1 mW, depending on what is being sensed.

Another critical component of an autonomously deployed edge device is its ability to transmit data to other proximal devices or a central location (hub) where the information can be analyzed. There are several communication modalities that could be used that operate near 1 V but differ in cost and distance range. A simple method would be to transmit the data optically by modulating an infrared light-emitting diode that operates at milliwatt power levels (but limited distance range). Vertical-cavity surface-emitting lasers are highly directional, providing longer range, but operate at several milliwatts. Yet another communication mode involves radio-frequency (RF) transmission being developed for ultralow-power operation in

IBM T. J. Watson Research Center, Yorktown Heights, NY 10598, USA. Email: rahaight@ibm.us.com

receive and transmit modes (6) with a total power consumption of 6 to 8 mW and data send and receive rates of 150 kilobytes per second (kbps) to 4 Mbps. Devices are larger for RF communication (typically centimeters or larger) because of the antenna size. To reduce edge device power needs, a hub system with a central high-power transmitter would only require low-power backscatter modulation from the edge device. Although the transmit power of the edge device is dramatically

stored charge is depleted. The use of batteries as a voltage regulator avoids the need for energy-consuming integrated-circuit voltage regulation. For a small computer-sensor device, we may consider a 1-cm<sup>2</sup> footprint for a battery that is 0.1 cm thick; the maximum energy stored at an energy density of 200 Wh/liter for a lithium-ion battery (7) is ~20 mWh. Given our estimates of power consumption for processing and data storage (~100  $\mu$ W), sensing (~1 mW), data transmis-

1% that of the required 300  $\mu$ m of Si necessary to fully absorb ambient light.

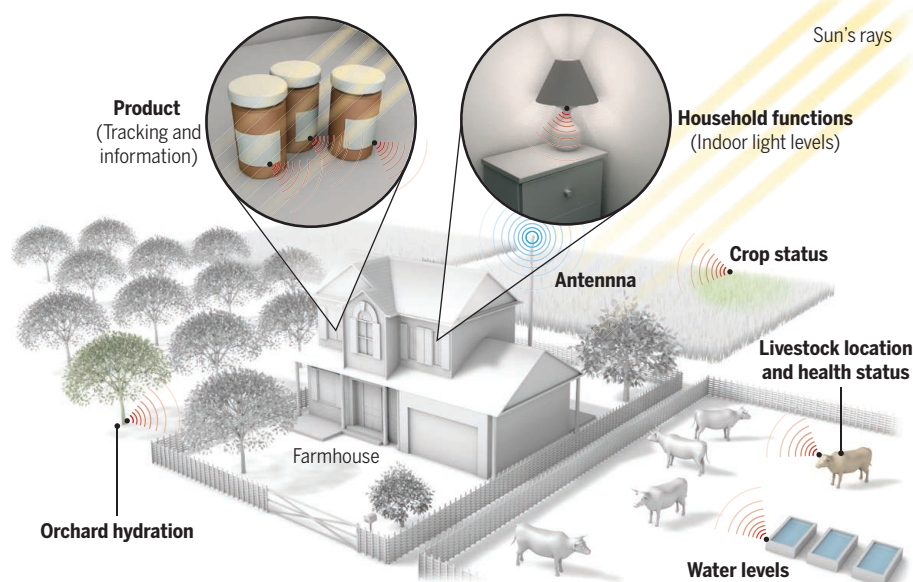
There are also other important considerations in choosing a thin-film PV device to be merged with a battery. The voltage produced by the PV device at its maximum power point ( $V_{\text{mpp}}$ ) must exceed that necessary to charge the battery. Although the standard voltages,  $V_{\text{oc}}$  (open-circuit voltage) and  $V_{\text{mpp}}$  of a PV device are typically determined under full-Sun illumination, called 1 sun (AM1.5) of ~100 mW cm<sup>-2</sup>, charging of the battery must occur under a wide range of illumination conditions and is a challenging aspect of PV-based energy harvesting. Typical fluorescent lighting produces about 10<sup>-3</sup> suns. At such low light levels, the shunt resistance (9) of the cell must be at least 500 to 1000 ohm-cm<sup>2</sup> to generate sufficient voltage to charge the battery.

Optimizing the band gap of the PV absorber with the light source is another key consideration. Indoor lighting from fluorescent or LED bulbs has a higher proportion of blue light than the Sun. The band gap of CZTS,Se can be correspondingly increased by increasing its sulfur content. Emerging alloys of CZTS,Se, including (Ag,Cu<sub>1-x</sub>)ZTS,Se (10) with a variable band gap and AgZTSe (11) with a 1.3-eV band gap, hold promise for even higher efficiency than traditional CZTS,Se because of suppression of bulk defects. The band gap tunability of CZTS,Se and CIGSe distinguishes them from Si, GaAs, and CdTe. Higher voltages can be achieved by connecting cells in series or fabricating a tandem cell in which a top PV cell with a larger band gap is deposited on a bottom cell of a smaller band gap (12).

Truly self-powered, intelligent computers and sensors provide the ability to communicate as a network and to gather information on critically important aspects of even the most remote regions of our planet. They also can help us to better control our nearby environment to make it more efficient and sustainable. ■

## Powering edge devices

The Internet of Things can connect sensors deployed remotely, but these sensors must be powered. Photovoltaics could power sensors both inside buildings and in the environment, as shown in this example of a "smart ranch."



reduced, the viable separation distance to its hub is limited.

Smart edge systems can also process data locally and minimize the amount of communication with the central system. The architecture of the network used to enable hub-edge device communication will dictate data rates and distance specifications. At present, ultralow-power devices (2) can transmit a few meters, requiring a series of other proximally located devices to relay data over longer distances. Data transmission over hundreds of meters or more is possible but incurs a trade-off between the power cost of processing and that for communication.

Given these power budget estimates, we focus on energy harvesting, voltage regulation, and energy storage. One attractive approach would be to merge thin-film PV devices with an energy storage medium such as a battery or supercapacitor. Batteries have a distinct advantage in that they deliver current at a constant voltage, determined by the particular electrochemistry used, until the

sion (~5 mW), and intermittent operation totaling 1 hour per day, such an ultrasmall battery would power an edge device for perhaps 3 days. In reality, systems deployed in the field would need to last a year or more.

The foregoing simple analysis reveals the acute need for energy harvesting to dramatically extend the life of the device in the field. Light, as a nearly ubiquitous source of energy, is an optimal choice. For PV cells, the light absorber needs to be cheap, efficient, and, perhaps most important for field applications, environmentally compatible, because once deployed, it likely will not be retrieved at a later date. The multielemental kesterite Cu<sub>2</sub>ZnSn(S<sub>1-x</sub>Se<sub>x</sub>)<sub>4</sub> (CZTS,Se) is the most efficient Earth-abundant thin-film absorber available today. Although its power conversion efficiency of 12.6% (8) is only a little more than half that of CdTe and Cd-InGaSe (CIGSe), it does not contain metals that are toxic (Cd) or expensive and relatively rare (In and Ga). The entire thin-film PV device stack is only 2 to 3  $\mu$ m thick, just

## REFERENCES

1. N. Gershenfeld et al., *Science* **327**, 1086 (2010).
2. P. Pannuto et al., in *Proceedings of the 42nd Annual International Symposium on Computer Architecture* (2015), pp. 629–641.
3. "ARM Launches Its Smallest, Lowest Power, Most Energy Efficient Processor" (press release); [www.arm.com/about/newsroom/24418.php](http://www.arm.com/about/newsroom/24418.php).
4. B. Zhai et al., *IEEE Trans. VLSI* **17**, 1127 (2009).
5. U. Guth et al., *Meas. Sci. Technol.* **20**, 042002 (2009).
6. IMEC, "IEEE 802.11AH Wi-Fi HaLow Radio in TSMC 40NM CMOS"; [www2.imec.be/content/user/File/Brochures/ip2016/7\\_ULP%20WiFi%20RADIO\\_mail.pdf](http://www2.imec.be/content/user/File/Brochures/ip2016/7_ULP%20WiFi%20RADIO_mail.pdf).
7. J. B. Goodenough, K. S. Park, *J. Am. Chem. Soc.* **135**, 1167 (2013).
8. W. Wang et al., *Adv. Energy Mater.* **4**, 130165 (2014).
9. T. J. McMahon et al., in *Proceedings of the 25th IEEE Photovoltaic Specialists Conference* (1996), pp. 1291–1294.
10. T. Gershon et al., *Adv. Energy Mater.* **6**, 1502468 (2016).
11. E. Chagarov et al., *J. Chem. Phys.* **144**, 104704 (2016).
12. J. Kim et al., *Science* **317**, 222 (2007).

10.1126/science.aag0476



## Solar-powering the Internet of Things

Richard Haight, Wilfried Haensch, and Daniel Friedman

*Science* **353** (6295), . DOI: 10.1126/science.aag0476

### View the article online

<https://www.science.org/doi/10.1126/science.aag0476>

### Permissions

<https://www.science.org/help/reprints-and-permissions>

Use of this article is subject to the [Terms of service](#)

---

*Science* (ISSN 1095-9203) is published by the American Association for the Advancement of Science. 1200 New York Avenue NW, Washington, DC 20005. The title *Science* is a registered trademark of AAAS.

Copyright © 2016, American Association for the Advancement of Science